- Presence of ethereal oils.
- Leaves are always simple net-veined.
- Each floral whorls with many parts.
- Perianth usually spirally arranged or parts in threes.
- Stamens with broad filaments.
- Anthers tetrasporangiate.
- Pollen monosulcate.
- Nectaries are rare.
- Carpels usually free and
- Embryo very small.

Monocots are classified into 11 orders and 77 families (basal monocots + lilioids + commelinids)

- Seeds with single cotyledon.
- Primary root short-lived.
- Single adaxial prophyll.
- Ethereal oils rarely present.
- Mostly herbaceous, absence of vascular cambium.
- Vascular bundles are scattered in the stem.
- Leaf simple with parallel-veined.
- Floral parts usually in threes.
- Perianth often composed of tepals.
- Pollen monosulcate.
- Styles normally hollow and
- Successive microsporogenesis.

Eudicots are divided into 45 orders and 313 families (early diverging eudicots + super rosids + super asterids).

- Seeds with always two cotyledons.
- Nodes trilacunar with three leaf traces.
- Stomata anomocytic.
- Ethereal oils rarely present.
- Woody or herbaceous plants.
- Leaves simple or compound, usually net-veined.
- Flower parts mostly in twos, fours or fives.
- Microsporogenesis simultaneous.
- Style solid and
- Pollen tricolpate.

APG system is an evolving system that might undergo change periodically based on the new sets of data from various disciplines of Botany. It is the currently accepted system across the world and followed by all the leading taxonomic institutions and practising taxonomists. However, it is yet to percolate into the Indian botanical curriculum.

Classification reflects the state of our knowledge at a given point of time. It will continue to change as we acquire new information.

Changes in earlier taxonomic understanding.

The newly proposed APG classification system has brought many changes in our earlier understanding on the concept of primitive flowering plant families. Some of them are given below:

- The real Ranalean families, especially the arborescent ones are no more the primitive families. But as per APG classification system, Amborellaceae, Nymphaceae, Austrobaileyaceae, Magnoliaceae and Chloranthaceae form the early angiosperms.
- Monocots are recognised as a monophyletic group and hence terminology is retained.
- Dicots are polyphyletic group and as a result the use of the term dicotyledons as a group becomes outdated.
- Liliaceae (Sensu lato) is split into 14 families.
- Molluginaceae and Gisekiaceae are recognised separately from Aizoaceae.
- Euphorbiaceae (s.l.) are split in to Phyllanthaceae, Picrodendraceae and Putranjivaceae.
- Asclepiadaceae are merged with Apocynaceae (s.l.)
- Many genera that were conventionally treated under Verbenaceae such as *Clerodendron*, *Tectona* and *Vitex* are transferred to Lamiaceae based on the modified circumscription of these families.

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